Please read the instructions below before the exam starts, and listen carefully to any instructions given by the invigilators. You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2023 **EC116 2022/3 A 800**

Two Answer books

UNIVERSITY OF READING

INTRODUCTORY MATHEMATICS FOR ECONOMICS 1 (EC116)

Two hours

Answer **ALL** questions (100 marks);

1. Consider the following model of the economy:

C = 200 + 0.6 (Y – T)
I = 300
G = 300
T = 0.2 Y

(a) Compute the equilibrium level of income of the economy.

(5 marks)

(b) At the equilibrium level of income, what is the value of the government budget surplus?

(5 marks)

(c) Sketch the consumption function, C, by using the slope-intercept approach.

(5 marks)

(d) Suppose G increases by 100, calculate its effect on equilibrium income.

(5 marks)

2. A monopolist's demand and total cost functions of a good are $Q = -P + 5$ and $C = Q + 2Q^2$, respectively.

(a) Derive the total revenue and profit (π) functions.

(6 marks)

(b) Sketch the profit function (for $Q \geq 0$ only; put π and Q on vertical and horizontal axis, respectively).

(6 marks)

(c) Calculate the profit maximising output level Q*. (Hint: verify the result with the second-order derivative)

(8 marks)

3. Given the following demand and supply functions $P = 400 - 0.5\mathrm{Q}$ and $\mathrm{P} = 200 + Q$, respectively,

(a) Calculate the equilibrium price and quantity in the market.

(8 marks)

(b) Sketch the demand curve, supply curve and your answers to part (a) in one diagram (for $P \geq 0$ and $Q \geq 0$ only; put P and Q on vertical and horizontal axes, respectively).

(8 marks)

(c) Calculate the price elasticities of demand and supply at the market equilibrium using differentiation.

(4 marks)

4. (a) Differentiate the following equations and find their slopes at x=2.

(i) $y = (x^3 + 2x)^2$
(ii) $y = ln(x^2 - x)$
(iii) $y = \frac{x}{e^{3x}}$
(iv) $y = (x^2 - x)(x^3 + 2)$

(16 marks)

(b) Find the stationary points of the following equation. Use second-order derivative method to check whether they are local maximum or minimum.

$$y = 2x^3 + x^2 + 2x$$

(8 marks)

(c) Given the demand function $Q = 10 - ln(3 + 2P)$, calculate the price elasticity of demand at P=2 by differentiation.

(8 marks)

(d) Use the rules of logs to expand each of the following:
(i) $\ln(xy)^3$
(ii) $\ln\sqrt{\frac{x^2y^2}{z}}$

(4 marks)

(e) Use the rules of logs to express each of the following as a single logarithm:
(i) 4ln x – 3ln z
(ii) ln a + 3 ln c – 2 ln b

(4 marks)

(End of Question Paper)